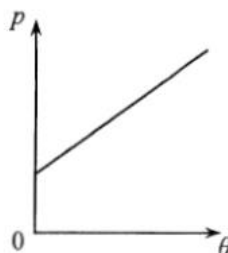
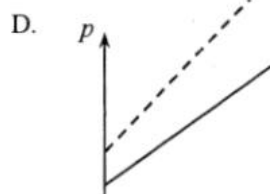
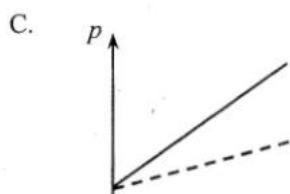
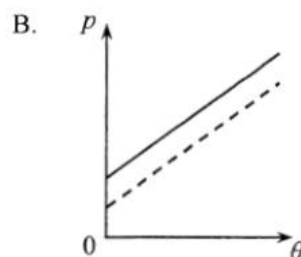
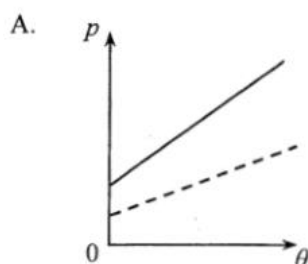


2012 Q3

- *3. An ideal gas is contained in a closed vessel of fixed volume. The graph below shows the variation of pressure p of the gas against its Celsius temperature θ .



If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ ?



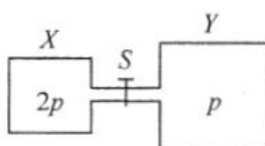
2013 Q3

- *3. In which of the following situations would the r.m.s. speed of the molecules of a fixed mass of an ideal gas increase?
- (1) The gas is heated under constant volume.
 - (2) The gas expands under constant pressure.
 - (3) The gas is compressed under constant temperature.

- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

2013 Q4

*4.



Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube as shown. Initially, tap S is closed and the same kind of ideal gas at the same temperature is contained in X and Y at pressure $2p$ and p respectively. The tap S is then opened and equilibrium state is finally reached with the temperature unchanged. Which statement is **INCORRECT** ?

- A. Before S is opened, both vessels contain the same number of gas molecules.
- B. Before S is opened, the average kinetic energy of the gas molecules in both vessels is the same.
- C. When S is opened, a net flow of gas from X to Y occurs.
- D. When equilibrium is reached, the gas pressure becomes $\frac{3}{2}p$.

2015 Q3

3. When an object P is in contact with another object Q , heat flows from P to Q . P must have a higher

- (1) temperature.
- (2) internal energy.
- (3) specific heat capacity.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

2016 Q3

- *3. When an ideal gas is heated from 25°C to 50°C , the average kinetic energy of the gas molecules will

- A. double.
- B. increase by 41 %.
- C. increase by 8.4%.
- D. increase by 4.1%.

2017 Q3

3. Which of the following statements about the internal energy of a substance are correct ?

- (1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
- (2) When a vapour condenses, its internal energy decreases.
- (3) When a liquid evaporates, the internal energy of the remaining liquid increases.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

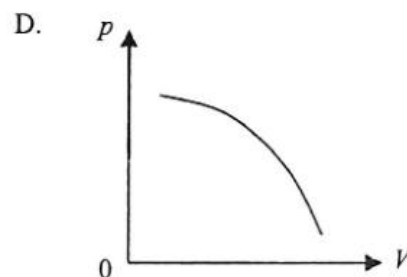
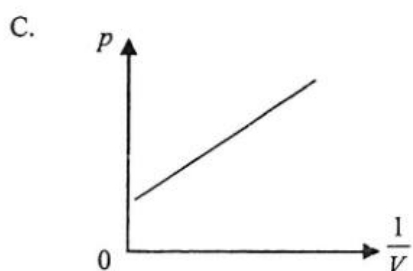
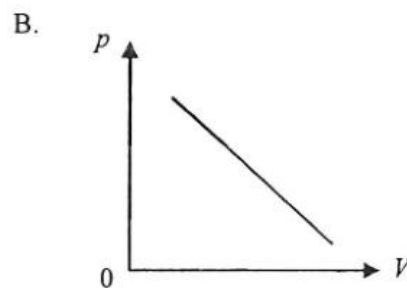
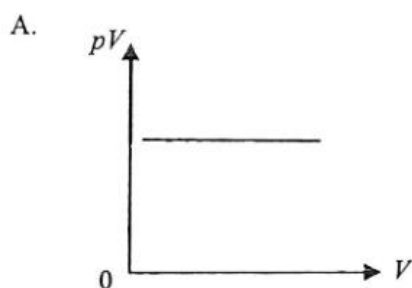
2017 Q4

- *4. The pressure of a fixed mass of an ideal gas at 10°C is $2 \times 10^5 \text{ N m}^{-2}$. If the volume of the gas is reduced to half of its original volume and its temperature is increased to 100°C , what would the pressure be ?

- A. $1.00 \times 10^5 \text{ N m}^{-2}$
- B. $1.32 \times 10^5 \text{ N m}^{-2}$
- C. $4.00 \times 10^5 \text{ N m}^{-2}$
- D. $5.27 \times 10^5 \text{ N m}^{-2}$

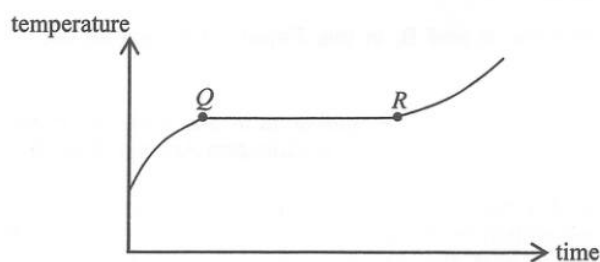
2018 Q4

- *4. From which graph below can one deduce that the pressure p of a fixed mass of an ideal gas is inverse proportional to its volume V when the temperature of the gas is kept constant ?



2019 Q2

2.



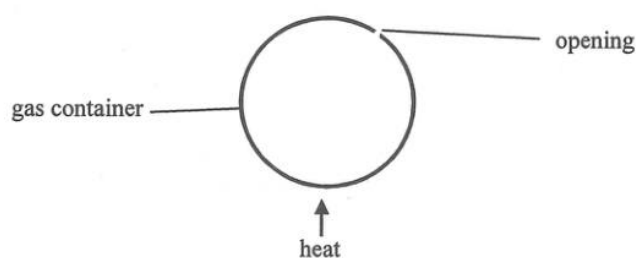
A substance undergoes the fusion process. The figure shows how the substance's temperature varies with time. Its temperature remains constant during the period from Q to R . Which of the following deductions **within this period** is/are correct ?

- (1) The substance does not absorb heat.
- (2) The mass ratio of the substance in solid and liquid states remains constant throughout.
- (3) The average potential energy of the molecules of the substance increases with time.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

2019 Q3

- *3. The figure shows an inexpandable container with an opening.



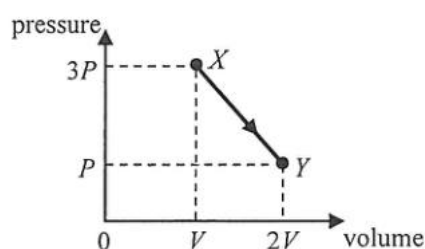
When the gas inside the container is being heated slowly by a heater, which statements about the gas molecules in the container are correct ?

- (1) The number of molecules decreases.
- (2) The average kinetic energy of molecules increases.
- (3) The average separation between molecules remains unchanged.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2020 Q3

- *3. A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure-volume graph below.



If the temperature of the gas at state Y is 25°C , what is its temperature at state X ?

- A. -74.3°C
- B. 16.7°C
- C. 37.5°C
- D. 174°C

2021 Q2

2. Metal blocks X and Y are identical in size and shape. X is of a higher temperature than Y . Which of the following statements must be correct ?

- (1) Energy will flow from X to Y if they are in thermal contact.
- (2) X is a better conductor of heat compared to Y .
- (3) The total internal energy of X is higher than that of Y .

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

2021 Q3

- *3. For an ideal gas, kinetic theory deduces that $pV = \frac{1}{3} Nmc^2$. Which physical quantity below can be represented by $\frac{3p}{c^2}$?

- A. the total mass of the gas
- B. the volume of one mole of the gas
- C. the density of the gas
- D. the number of gas molecules per unit volume

2022 Q2

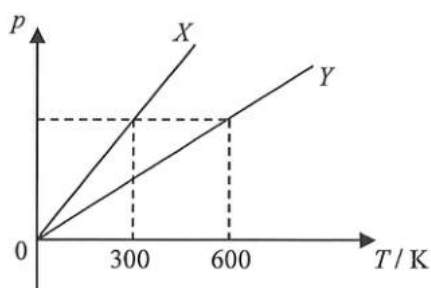
2. When ice melts into water at 0°C , what happens to the molecules during the melting process ?
- (1) their average separation increases
 - (2) their average kinetic energy increases
 - (3) their average potential energy increases
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

2022 Q3

- *3. A weather balloon filled with 21 kg of helium gas has a volume of 120 m^3 at 27°C . Find the gas pressure in the balloon. Given: mass of one mole of helium gas = $4 \times 10^{-3}\text{ kg}$
- A. $9.93 \times 10^4\text{ Pa}$
 - B. $1.00 \times 10^5\text{ Pa}$
 - C. $1.05 \times 10^5\text{ Pa}$
 - D. $1.09 \times 10^5\text{ Pa}$

2023 Q3

- *3. A fixed mass of an ideal gas contained in a closed vessel X of volume V is heated. Line X in the figure shows the variation of pressure p with temperature T of the gas. The experiment is repeated with another closed vessel Y containing the same amount of this ideal gas and line Y represents the result.



Find the volume of vessel Y .

- A. $0.25 V$
- B. $0.5 V$
- C. $2V$
- D. $4V$

Book 1 section 4 Gas Law and Kinetic Theory: DSE multiple choice

2024 Q3

- *3. The best vacuum attained in the laboratory has a pressure of about 10^{-8} Pa. Estimate the order of magnitude of the number of air molecules in 1 cm^3 of such 'vacuum' at room temperature.

- A. 10^4
- B. 10^6
- C. 10^8
- D. 10^{12}

2025 Q3

- *3. Identical closed vessels *X* and *Y* contain carbon dioxide (CO_2) and oxygen (O_2) respectively.



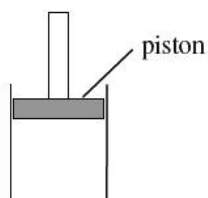
Both gases are at room temperature and atmospheric pressure. Which of the following physical quantities of the gas molecules in the two vessels is/are the same ?

- (1) root-mean-square speed
- (2) average kinetic energy
- (3) frequency of collision with the walls of the vessel

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

PP Q5

- *5. A fixed mass of an ideal gas is contained in a cylinder fitted with a frictionless piston as shown in the figure below. If the gas is cooled under constant pressure,



- (1) the average separation of the gas molecules will decrease.
- (2) the r.m.s. speed of the gas molecules will decrease.
- (3) the number of collisions per second of the gas molecules on the piston will decrease.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)